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How Physical Activity Context Relates to Cognition Across the Lifespan: A Systematic Review and Meta-Analysis

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Although numerous reviews and meta-analyses have examined the effects of physical activity on cognition, no previous meta-analysis has comprehensively explored the role of contextual factors. In this systematic review, we examined the role of contextual moderators along with more commonly investigated individual (e.g., age) and physical activity-related moderators (e.g., intensity, type). A multilevel meta-analysis was applied to 171 chronic (654 effect sizes) and 68 acute studies (305 effect sizes) involving 48,625 participants from preschool to older adulthood. On average, small positive effects on cognition were found for participation in both chronic ($g = 0.25$, 95% CI [0.19, 0.31]) and acute physical activity ($g = 0.21$, 95% CI [0.12, 0.30]). Physical activity effects on cognition seem to be jointly moderated by contextual and physical activity-related factors. The largest effect sizes were depicted for chronic practice of outdoor physical activity of moderate-to-vigorous intensity ($g = 0.63$, 95% CI [0.41; 0.85]) and with high cognitive demand ($g = 0.53$, 95% CI [0.15, 0.91]), and for acute bouts of outdoor physical activity of moderate intensity ($g = 0.71$, 95% CI [0.42, 0.99]). These results may inform the design of future chronic and acute physical activity trials to foster a more comprehensive understanding of the intersection between multiple moderators.

Public Significance Statement

This meta-analysis highlights the complex and multifaceted nature of the relation between physical activity and cognition across the lifespan, suggesting that the context, intensity, and cognitive demand of physical activity jointly matter. From a public health standpoint, emphasizing the commonly overlooked role of context beyond that of a mere background of physical activity interventions represents progress in identifying the contexts (e.g., outdoor physical environment) that, in conjunction with specific physical activity characteristics (e.g., moderate-to-vigorous intensity and high cognitive demand), are best suited to promote cognitive health across the lifespan.

Keywords: environment, physical activity type, dose, chronic, acute

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Myrto F. Mavilidi played a lead role in conceptualization, investigation, methodology, writing–original draft, and writing–review and editing.

continued

The number of studies examining the effects of physical activity on cognition in children (Álvarez-Bueno et al., 2017), adolescents (Esteban-Cornejo et al., 2015; D. R. Lubans et al., 2021), and adults (de Souto Barreto et al., 2016; Etnier et al., 2019) has accelerated rapidly in the last 10 years. Cognition refers to the processes, mental structures, and operations that facilitate perception, memory, and action (Posner, 1973). Physical activity and cognition research have especially concentrated on higher level cognition responsible for goal-oriented actions and behavioral flexibility (e.g., executive function; Diamond, 2013).

Accumulating evidence that physical activity benefits cognition, particularly in children, older adults, and special populations with suboptimal cognitive functioning, led to an authoritative position statement (Erickson et al., 2019). In contrast, other reviews have suggested that the available evidence does not support this notion at various ages (e.g., Cox et al., 2016; Singh et al., 2019). Two recent meta-reviews across the lifespan have attempted to clarify this issue, focusing on randomized controlled trials (RCTs) to address the causality of chronic (i.e., long-term practice of) physical activity on cognition in healthy individuals, but using different methodological approaches (Ciria et al., 2023; Pesce et al., 2023).

On the one hand, Ciria et al. (2023) performed a quantitative umbrella review of meta-analyses in which the outcomes of each meta-analysis were reanalyzed using a multilevel modeling approach that corrected for risk of publication bias. Although most of the included meta-analyses reported a small positive effect of physical activity on cognition, these effects were no longer significant once bias correction was applied. Notably, this umbrella review included only "healthy participants" and purely "physical" activity interventions, using relatively stringent selection criteria. Studies with participants who had not developed diseases or syndromes but were at risk of developing them (e.g., overweight/obese population, with rates of 42% among the United States, 32% Australian, 28% Canadian, 27% British adults; World Obesity, 2024) as well as interventions that entailed cognitively demanding physical activities (Tomporowski & Pesce, 2019) or mind–body holistic movement practices (Vergeer & Biddle, 2021) were excluded. Thus, Ciria et al.'s relatively narrow inclusion criteria might limit the generalizability of the results in an otherwise methodologically strong meta-analysis.

On the other hand, Pesce et al. (2023) adopted both an empiricist and constructivist lens by performing a qualitative umbrella review of systematic reviews and meta-analyses, and a theory-driven exploration of the underlying causal mechanisms (i.e., realist synthesis; Hunter et al., 2022), respectively. The umbrella review highlighted an inconsistent grading of methodological quality, an overlap of reviews with partially similar pools of primary studies, and the role of several moderators that may be responsible for the inconsistent results across reviews.

Similar issues emerge from reviews focused on the acute effects of physical activity (i.e., transient effects of a single physical activity bout) on cognition in children (Hillman et al., 2019), adults (Zheng et al., 2021), older adults (McSween et al., 2019), and across the lifespan (Ishihara et al., 2021). These reviews highlight methodological, sampling, and individual difference issues that may explain the variability of the findings. Examples of issues include differences in the assessment of cognitive outcomes (Hillman et al., 2019; Pontifex et al., 2019) and physical activity dose and duration (Chang et al., 2012) as well as individual differences in cognition, measured at pretest and meta-analyzed at the individual participant level. Accounting for individual assessments at baseline allows for the identification of potential larger gains in executive function for lower performers at baseline (Ishihara et al., 2021).

A range of psychosocial, behavioral, and neurobiological mechanisms may explain the effects of chronic and acute physical activity on cognition (D. Lubans et al., 2016, 2021; J. Marmeleira, 2013). Some accounts suggest that effects are mostly driven by quantitative characteristics (e.g., dose and intensity), which impact brain regions involved in cognitive function, motor planning, and control (e.g., frontal cortex, hippocampus, cerebellum, basal ganglia). The neurotrophic stimulation hypothesis suggests that chronic participation in physical activity leads to elevated circulating levels of neurotrophic factors and overall cerebrovascular function independently of or relatedly to improvements in fitness (Erickson et al., 2019; E. I. Walsh et al., 2020). In this latter case, the neurotrophic stimulation hypothesis is complemented by the cardiovascular fitness hypothesis (Agbangla et al., 2021). The neurotrophic stimulation resulting from participation in physical activity contributes to improvements in brain structure and function (Valkenborghs et al., 2019).

Similarly, enhanced levels of neurotrophic factors may also explain the acute effects of physical activity on cognition (Piepmeier & Etnier, 2015). When physical activity has the features of a single exercise bout, its transient effects on the brain and cognition may vary depending on the intensity and duration of the physical activity. These effects may trigger beneficial or detrimental changes in brain biochemistry (catecholamine hypothesis; McMorris, 2016) depending on whether the cognitive assessment is successive or concomitant to the physical activity bout (Tomporowski & Qazi, 2020). Detrimental effects of high-intensity exercise performed concomitantly to an executive function task are interpreted as competition for limited

Spyridoula Vazou played an equal role in conceptualization, data curation, investigation, methodology, and writing–review and editing. David R. Lubans played an equal role in conceptualization, formal analysis, methodology, supervision, and writing–review and editing. Katie Robinson played a supporting role in data curation, formal analysis, methodology, and visualization. Andrew J. Woods played a supporting role in data curation and methodology. Valentin Benzing played a supporting role in data curation and methodology. Sofia Anzeneder played a supporting role in data curation and methodology. Katherine B. Owen played a lead role in formal analysis, supervision, and validation, a supporting role in methodology, and an equal role in data curation. Celia Álvarez-Bueno played a lead role in formal analysis. Levi Wade played a supporting role in data curation, formal analysis, and methodology. Jade Burley played a supporting role in data curation and methodology. George Thomas played a supporting role in data curation and methodology. Anthony D. Okely played a supporting role in data curation and methodology. Caterina Pesce played a lead role in investigation, methodology, and supervision and an equal role in conceptualization, writing–original draft, and writing–review and editing.

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resources in the brain areas subserving those functions (i.e., hypofrontality hypothesis; Dietrich, 2003; Jung et al., 2022).

More generally, acute exercise effects on cognition have been explained by referring to the theory of physiological arousal, according to which arousal and cognitive performance are linked by an inverted *U*-function (McMorris & Hale, 2015). Nonetheless, the application of this theory to the cognitive effects of arousing exercise bouts has mainly received disconfirmation. Meta-analytically, cognitive enhancements emerge for aerobic physical activity already at or above the light-intensity threshold (Lambourne & Tomporowski, 2010). A positive effect, though small, was also observed when exercise intensity was set high (e.g., in high-intensity interval training), regardless of its mainly short duration (Moreau & Chou, 2019). Instead, hard-to-very-hard physical activity seems to be beneficial to cognitive function after a delay following exercise but not immediately after it (Chang et al., 2012).

Further mechanisms have been suggested to underlie the effects of the qualitative characteristics of chronic and acute physical activity on cognition. Qualitative characteristics can be regarded as the degree of cognitive and coordinative challenge due to physical activity, task novelty and complexity, or variability of its situational conditions. The expected impact on cognition of this kind of challenge is referred to as the cognitive stimulation hypothesis (Pesce, 2012). This is the case, for example, of the different cognitive and motor coordination demands of simple motor tasks such as riding a stationary bike versus physical activities that involve complex cognitive demands (e.g., learning a dance choreography).

Positive cognitive outcomes based on the physical activity type were found after both prolonged programs and acute bouts of physical activity for children and adolescents (e.g., Paschen et al., 2019; Pesce, Ballester, & Benzing, 2021; Vazou et al., 2019) and adults (Levin et al., 2017). For example, during developmental stages, working memory seems to be affected differently by coordinative and aerobic exercise training (Ludyga et al., 2022). Despite the largest benefits emerge for coordinative exercise, likely because of its motor skill acquisition component (Tomporowski & Pesce, 2019), aerobic exercise also benefits working memory but only when the intensity of exercise is vigorous. Regardless of the fitness level, aerobic exercise seems to be beneficial for executive functions, with the largest effects observed in preadolescent children and older adults (Ludyga et al., 2016). In the aging population, different physical activity types seem to elicit clinically relevant effects on general cognitive function at different doses, already starting at a low dose (Gallardo-Gómez et al., 2022). Predicted maximal significance was found with a nonlinear positive relationship at different metabolic equivalents (METs)-min/week for aerobic and resistance exercises compared to body and free weights and machines, mixed aerobic exercises, and walking. Furthermore, cognitively enriched physical activity interventions that involve skill acquisition demands may elicit greater cognitive benefits than aerobic exercise (Tomporowski & Pesce, 2019). An integrative view applied to physical activity in educational settings (M. F. Mavilidi et al., 2022) suggests that the cognitive outcomes of different combinations of motor and cognitive task demands depend on the level of temporal connection and content relatedness (e.g., highest memory performance during physically active lessons; M. F. Mavilidi, Ruiter, et al., 2018; M. F. Mavilidi et al., 2022).

Interestingly, after a flourishing debate on the characteristics of physical activity that can influence its efficacy for improving cognition (e.g., Diamond & Ling, 2016, 2019; Hillman et al., 2019), and after the publication of meta-analytic evidence that both quantitative and qualitative characteristics of physical activity may act, independently or jointly, as moderators of physical activity effects on cognition (Ludyga et al., 2020), a renewed debate emerged (Ciria et al., 2023, 2024; Dupuy et al., 2024). The research work that has triggered this new debate (Ciria et al., 2023) has a methodological lens that focuses on the initial fundamental question of the existence of overall physical activity effects on cognition, shifting the focus away from the nuances of the interplay of moderators. To pave a new way forward in the analysis of moderators and mediating mechanisms, Pesce et al.'s (2023) meta-review extended the focus from the more commonly considered quantitative and qualitative characteristics of physical activity to the largely neglected role of the physical activity context. Following that work, we conceive the context of physical activity interventions not merely as the background of the implementation but, most importantly, as a facilitator or constraint (Coles et al., 2017) and a trigger of generative mechanisms (de Souza, 2013) of physical activity effects on cognition.

Pesce et al.'s (2023) evidence synthesis highlighted that cognitive capacity could be "boosted" through mechanisms triggered by energetic features of the dosage of physical activity (e.g., Chang et al., 2012) and "built" through mechanisms of cognitive training by cognitively demanding physical activity types (i.e., skill acquisition hypothesis, Tomporowski & Pesce, 2019), but also "restored" through mechanisms triggered by features of the physical activity context (e.g., green environment hypothesis, Stevenson et al., 2018). There is increasing interest in the physical activity performed in the natural environment ("green exercise") for the expected additive physical and psychological benefits that may be accrued by exposure to nature and physical activity (Baena-Morales et al., 2023; Bamberg et al., 2018). Cognitive benefits have been explained (de Keijzer et al., 2016) by suggesting that exposure to outdoor natural environments allows for the restoration of top-down attention resources (attention restoration theory; Kaplan, 1995). This hypothesis is mainly grounded on evidence of transient beneficial effects of nature exposure that likely allow recharging attention resources (McDonnell & Strayer, 2024). Another not mutually exclusive theory suggests that exposure to nature may transiently restore these resources by reducing negative factors that impair them, such as stress (stress reduction theory; Ulrich et al., 1991). The search for empirical support for this theory is still ongoing (Mygind et al., 2021).

Exploring the role of contextual factors may help reconcile increasing divergences in conclusions of reviews and meta-analyses by explaining why physical activity interventions that involve similar doses or types of physical activity sometimes yield inconsistent results (Pesce et al., 2023). Complementing the view on quantitative and qualitative physical activity characteristics with a focus on the context may provide practitioners and policymakers with valuable information on what works (quantity/quality; e.g., physical activity dose or type) for whom (individual; e.g., age), under which circumstances (i.e., context) and why (i.e., underlying mechanisms). These questions are addressed within different theoretical frameworks (e.g., Pawson et al., 2005; Rothman & Sheeran, 2021) or mixed data-driven and theory-based approaches (Pesce et al., 2023).

Therefore, our current review aimed to quantify the moderating role of contextual factors of chronic and acute physical activity on cognition across the lifespan. We conceptualized contextual factors as (a) physical environment, (b) social environment, (c) experimental domain, (d) delivery mode, and (e) delivery style. We also explored the potential interactive influence of physical activity tasks and context on cognition. Thus, the following research questions were addressed:

1. *Primary*: Do contextual factors moderate the effects of chronic and acute physical activity on overall cognition?
2. *Exploratory*: Do physical activity tasks (quantity and quality) and contextual factors interact as moderators of cognitive outcomes?

Method

Search Strategy

We conducted a structured electronic search of all publication years until August 2023 using ERIC (ProQuest), APA PsycInfo, PubMed, Scopus, and SPORTDiscus with full text. We did not put any limitations on the language of the search (we used Google Translate during screening and contacted authors for information, when there was a potential match with the inclusion criteria). First, titles and abstracts identified in the search were screened for suitability. Second, full-text reports were screened against the study selection criteria. Finally, reference lists from full-text reports were searched to identify any additional publications not found in the initial search (see Supplemental File 1 for search terms). Differences in the extraction and coding of information among assessors who performed the screening process (MFM, SV, CP, SA, VB, AO, AW, GT) were discussed and resolved separately by assessors (CP, SV, DL).

Selection Criteria

Inclusion Criteria

1. Studies that present original data from chronic or acute physical activity interventions: The distinction was made depending on whether the study assessed the longer lasting effects of an intervention with repeated physical activity bouts (of at least 4 weeks duration) or the transient effects of a single bout of physical activity assessed immediately and/or at different time points after physical activity bout cessation (minutes, hours).[1] To classify interventions with multiple physical activity bouts as chronic or acute, the distinguishing factor was whether the aftereffects on cognition were assessed only once at the end of the overall program period (i.e., chronic effects) or after each physical activity bouts (i.e., acute effects).
2. For both chronic and acute physical activity studies: experimental studies with random assignment, between-subjects design, or group/cluster RCTs with both pre- and posttest data.
3. For acute studies, physical activity only: Randomized within-subjects crossover trials (Cochrane Handbook for Systematic Reviews of Interventions, 2023) with pre–post or posttest-only design (Pontifex et al., 2019), with or without counterbalancing, and a washout period not shorter than 2 days to avoid carryover effects that have not been reported for intervals of 48 hr and beyond (Pontifex et al., 2019; van Dongen et al., 2016). Given the cognitive test repetitions within a short time, the article should contain information regarding the robustness of the test against learning effects or the use of a sufficiently long cognitive test training.
4. Studies in preschool children (3–5 years), children (6–12 years), adolescents (13–17 years), adults (18–59 years), and older adults (60+ years).
5. Population with typical development (including overweight and obese).
6. Passive, nonphysically active, and/or physically active control conditions.
7. Studies assessing executive function (inhibitory control, working memory, cognitive flexibility), memory, attention, and intelligence, with outcome variables including only standardized assessments.
8. Neuroscience studies including a behavioral assessment of cognitive function.
9. Studies with physical activity tasks involving gross motor skills.
10. Published articles and gray literature (i.e., dissertations, fully published conference proceedings).

Exclusion Criteria

1. Qualitative studies, case studies or practice reports, observational studies.
2. For chronic physical activity studies: quasiexperimental design with no control group or posttest-only design (without pretest assessment).
3. For acute physical activity studies: within-subjects design without crossover or with a nonrandomized crossover or with cognitive task performance concomitant to physical activity (i.e., in-task acute physical activity studies).
4. Studies with children younger than 3 years.
5. Participants from special populations with clinical conditions (e.g., diseases or disabilities).
6. Studies with multiple experimental conditions, but none of them explicitly indicated a control group (e.g.,

[1] The authors thank an anonymous reviewer for their deep and generous engagement in providing relevant feedback, especially but not only in regard to the methodological approach to the selection and data extraction from acute studies.

resistance vs. yoga; different physical activity intensities/durations).

7. Studies including only health-related outcomes (e.g., fitness, weight status).
8. Studies including only neuroimaging/neurophysiological measures, academic achievement (generally limited to school-age populations) and cognitive outcomes using subjective rating scales, nonstandardized assessments, or on-task behavior.
9. Studies with physical activity tasks only involving fine motor skills or stationary physical engagement through alteration of environmental design (e.g., standing desks).
10. Multicomponent interventions with additional components to physical activity (e.g., physical activity and nutrition or followed by computerized cognitive training).
11. Studies with no relevant descriptive statistics (mean and standard deviation or standard error or confidence interval and sample size per group) provided for meta-analyses.

Selection of Moderators

An effect modifier (or moderator) is a variable that influences the strength of the relationship between a program and outcomes (Bauman et al., 2002, see Footnote 1). We examined a range of potential moderators relating to physical activity task and contextual characteristics (hereafter referred to as task- and context-level moderators, respectively). We also examined individual-level moderators (based on participant characteristics) and a range of study-level moderators (e.g., comparator, implementation fidelity). Some moderators were only included in the systematic review but not in the meta-analysis when there were not enough studies for the meta-analysis or when they covaried with other moderator categories of major interest for this review. For example, experimental setting was not included in the meta-analysis, because some categories covaried with those of the experimental domain (e.g., school is both a setting and a daily life domain). A map of the interplay between contextual, individual- and task-level moderators is presented in Figure 1 (adapted from Pesce et al., 2023; Vella, Aidman, et al., 2023).

Context-Level Moderators

This refers to the external and situational conditions in which programs occurred. The different categories and subcategories are presented in Table 1.

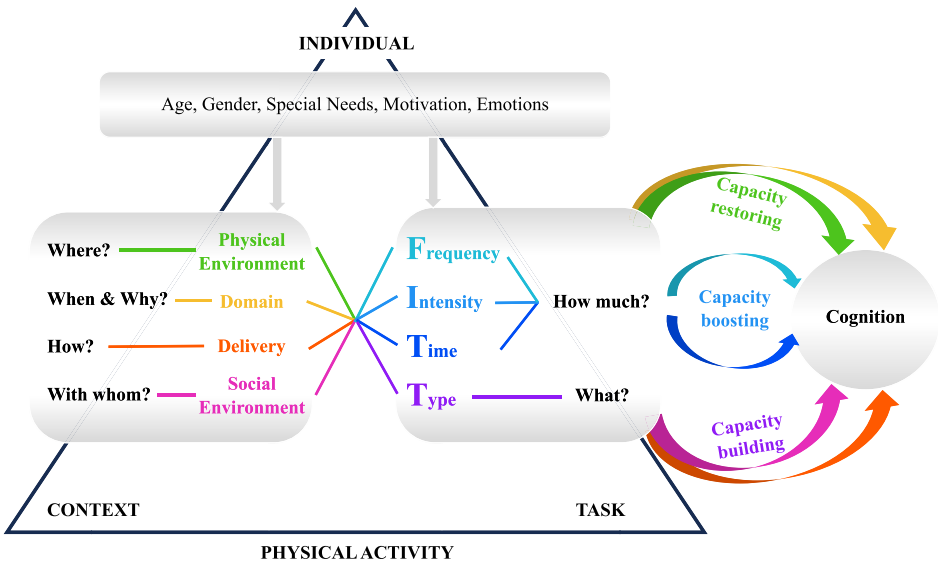
Physical Environment. This category refers to the location of the physical activity (i.e., where you do physical activity), including physical activity conducted indoors (e.g., gym, hall, classroom) or outdoors. A green environment seems to have positive psychosocial (cognitive, emotional) and physiological outcomes (better sleep, less sedentary behavior; Mnich et al., 2019).

Social Environment. This category refers to social engagement (i.e., who do you do physical activity with) involving physical activity conducted in groups or individually. According to Diamond (2012), unmet emotional or social needs work against displaying good executive functions, whereas social support may favor them.

Experimental Setting. This category refers to the organizational structure in which the physical activity occurred. The most common setting found in youth populations was school, whereas for

Figure 1

Physical Activity Contextual Factors That, Independently or Interactively With Task and Individual Factors, May Moderate the Effects of Physical Activity on Cognition



Note. Mechanisms of capacity boosting, building, and restoring are also hypothesized to underlie the physical activity effects on cognition. Inspired by Vella, Sutcliffe, et al. (2023) and Tomporowski and Pesce (2019). See the online article for the color version of this figure.

the older population, community sites (indoor, e.g., gymnasium; and outdoor, e.g., park), home, and laboratory/university were predominant (Pesce et al., 2023).

Experimental Domain. This category refers to the domain of life in which physical activity occurred (i.e., when and why you do physical activity), including school, leisure, commuting, and work. School sport, leisure time, and commuting, but not work-related physical activity, were identified as the main domain-specific characteristics that may explain the relationship between physical activity and mental health (White et al., 2017).

Delivery Mode. This category refers to the conditions under which the program was delivered (i.e., how you do physical activity), including face-to-face, remote, or virtual reality delivery modes. There seems to be no differences in participants' cardiorespiratory fitness outcomes in face-to-face compared to remote delivery physical activity interventions in adults (Richards et al., 2013), but positive outcomes for remote and web interventions compared to minimal or no interventions (Foster et al., 2013; A. M. Müller & Khoo, 2014). This category, originally not included in the International Prospective Register of Systematic Reviews (PROSPERO) protocol, was added to account for the feature of delivery that did not fall within the planned "delivery style" category.

Delivery Style. This category refers to the instructional style or quality (i.e., the way) physical activity was delivered (i.e., whether and where the delivery quality is grounded; Vella, Aidman, et al., 2023). For our review, we classified studies depending on whether the program followed a theory-based, top-down approach (e.g., grounded on self-determination theory; Ryan & Deci, 2000); a bottom-up, evidence-based approach, grounded on evidence of efficacy of specific strategies in supporting physical activity engagement and/or outcomes (e.g., using music; Terry et al., 2020); or an atheoretical approach. We followed a hierarchical order based on importance (theory-based, evidence-based, atheoretical). When two options were available (e.g., theory-based and evidence-based; e.g., D. R. Lubans, Leahy, et al., 2021), we chose the highest in the hierarchy (i.e., theory-based).

Physical Activity-Level Moderators

Quantitative and qualitative characteristics of the physical activity itself were also coded.

Intensity. Physical activity intensity is typically operationalized as light, moderate, moderate-to-vigorous, vigorous, and near-maximal intensity. These intensity levels were classified (Norton et al., 2010) based on threshold values of METs, %maximum heart rate (%HRmax), %Heart Rate Reserve, %maximum oxygen consumption (i.e., objective measures), and Rating of Perceived Exertion (C and C-R; i.e., subjective measures) which vary by age (e.g., Centers for Disease Control, 2022; Liguori & American College of Sports Medicine, 2021). For example, higher heart rate (HR) and MET thresholds have been recommended and used in pediatric physical activity literature for moderate-to-vigorous physical activity (MVPA; HR > 150 bpm) or vigorous activity (7 METs) due to higher resting HR and metabolic rate in children, whereas vigorous intensity in adulthood corresponds to METs $\geq$ 6 for younger and middle-aged adults and METs $\geq$ 4.8 for older adults. When no quantitative measure but only a qualitative description of the activities were provided, age-related compendia of physical activities (Butte et al., 2018; Herrmann et al., 2024; Willis et al., 2024) were used to estimate its energy cost (METs) that, in turn, was used to classify the intensity level.

To evaluate the type of assessment of physical activity intensity, we added the following category, which was defined based on the American College of Sports Medicine Guidelines for Exercise Testing and Prescription (2024): absolute versus relative intensity assessments. For absolute intensity assessments, for example, we considered accelerometry, HR, MET and also referred to conventional age thresholds (i.e., absolute intensity by age; Liguori & American College of Sports Medicine, 2021). As relative intensity assessments, for example, we included, for all ages: % HR_{max}, where HR_{max} = [220 − age] (or further adapted formulas including a coefficient, as [208 − 0.7 × Age] for children or [0.6 × (220 − Age)] for younger adults or [206.9 − 0.67 × age] for older adults); %Heart Rate Reserve, computed as percentage of the difference between observed resting HR and estimated (children and adults) or assessed (adults only, gold standard) HRmax; and self-perceptions (e.g., Rating of Perceived Exertion).

Time (or Duration). Among other moderators, program duration has been found to predict the effect size of exercise on cognitive function in healthy individuals, with interventions lasting longer proving to be more effective with long-session duration (Ludyga et al., 2020).

Type. Differentiating physical activity types is relevant to highlight whether cardiovascular fitness (North et al., 1990), cognitive stimulation (Pesce, 2012), or both hypotheses are predominant. Categories were slightly adapted to integrate interventions that, though fulfilling the inclusion criteria, were hardly categorizable as planned. Resistance exercise has been found to be superior to other types of exercise on cognitive function in older adults (Gallardo-Gómez et al., 2022), whereas across the lifespan, coordinative exercise seems to be most beneficial (Ludyga et al., 2020). Exergames have also emerged as a potential means for improving cognitive functions in adults (Gallou-Guyot et al., 2020; Ogawa et al., 2016) and children (Best, 2012; Staiano & Calvert, 2011). Holistic movement practices that unify physical activity and mindful movements (Vergeer & Biddle, 2021; Vergeer et al., 2021) are emerging as a highly consistent determinant of cognitive benefits (Diamond & Ling, 2020).

Level of Cognitive Demand. Cognitive engagement is related to the degree of attention and cognitive effort necessary during the learning of complex motor skills and execution of complex physical activity tasks (Pesce, 2012). In "cognitively demanding physical activity," cognitive-energetic resources "freed" by physical activity are likely allocated through mental effort in cognitive performance (Pesce, 2012). For our review, we have classified the cognitive demand in physical activity as low, high, or hybrid.

- Low: (a) repetitive exercise and (b) cyclic sports, involving automatized and cyclic movement patterns as running and cycling, or blocked movement repetitions as in circuit training.
- High: (a) activities explicitly designed to train cognitive function (i.e., cognitively engaging physical activity); (b) motor activities with explicit focus on motor learning and coordinative task complexity (i.e., assuming that when learning new skills, participants are involved in the initial

Table 1
Summary of Moderators (Context-, Physical Activity-, and Study-Level Moderators)

General category	Subcategory	Category in results	Explanation/example
Context-level moderators			
Environment	Physical	Indoors	Hall, gym
		Outdoors	Park, in nature, green environment
		Combined	Combination of indoor and outdoor
		No information	No information provided
	Social	Individual	Alone, individually
		Group	In groups or teams
		Combined	Combination of individual and group activities
Experimental	Setting	Laboratory/university	Controlled lab settings and university facilities
		School	School PE, recess, classroom-based, childcare
		Community	Elderly and fitness centers; retirement, age care and prevention settings; parks
		Home	Ecological home setting
		Combined	Combination of different settings
	Domain[a]	School	School-based time
		Leisure	Free time
		Occupational	Work time
Delivery	Mode	Face-to-face	Delivered by experts/teachers
		Virtual reality	Delivered by means of virtual-reality technique
		Remote	Non-face-to-face, distance (online) or self-delivery
		Combined	Multimodal or joint face-to-face and remote
	Style	Theory-based	Built as an operationalization of theoretical principles (e.g., self-determination theory)
		Evidence-based	Based on scientific efficacy evidence of efficacy (e.g., music)
		Atheoretical	Without reference to theory or scientific evidence
Physical activity-level moderators			
Quantitative characteristics	Program duration	(chronic studies only)	Number of weeks a program was implemented
	Session duration	(acute studies only)	Number of days a program was implemented
	PA intensity	Light physical activity	Light PA
		MPA	Moderate PA
		MVPA	Moderate-to-vigorous PA
		VPA	Vigorous PA
		Near-maximal PA	Near-maximal PA
	PA intensity (relative/absolute)	Relative	% age-adjusted heart rate max, % Heart rate reserve
		Absolute	Accelerometers
		Both	
		No information	No information provided
	Type of PA intensity measure[a]	Subjective rating	Ratings of perceived exertion
		Objective assessment	Accelerometers, heart rate monitors
		Both	Combination of subjective and objective assessment
		Author suggestion	Intensity reported by authors but not objectively measured
		Expert judgment	Intensity estimated by authors of this review from PA type reported and matched to metabolic equivalents based on published age-related compendia of energy cost of physical activities
Qualitative characteristics	PA type	Aerobic	Running, cycling
		Resistance	Squats, leg press
		Coordination	Fundamental movement skill interventions, dance
		Multicomponent motor–cognitive activities/ games	Combination of different PA types or exercise training; PE Motor–cognitive dual-task training, physically active learning, cognitively engaging activities, traditional physical activity games, exergaming combat sports, and team sport games

(*table continues*)

Table 1 (*continued*)

General category	Subcategory	Category in results	Explanation/example
		Holistic movement practices	Mindful movements (e.g., Yoga, QiGong)
	Cognitive demand	Low	Repetitive exercise, cyclic sports (involving automatized and cyclic movement patterns as running and cycling, or movement repetitions as in circuit training)
		High	Activities explicitly designed to training cognitive functions or learning new or complex motor skills; motor–cognitive dual-task (DT) training; open skill sports (team sport games, martial arts); exergaming with variable perceptual–motor demands
		Hybrid	Activities not explicitly designed to include cognitive challenge and/or coordinative complexity, and/or DT training; sports in between the closed and open skill continuum, without time pressure and an explicit focus on coordinative complexity
		No information	No information provided
Study-level moderators			
Implementation fidelity	Adherence (chronic studies only)	As prescribed	Average attendance rate of at least 80% of sessions
		Somewhat as prescribed	Average attendance rate lower than 80%
		No information	No information provided, or very large range (e.g., 10%–90%) with no indication of prevalent attendance range
	Compliance	Full researcher control	Full researcher control with manipulation checks
		Partial researcher control	Researcher-led without manipulation checks
		Modest researcher control	Specialist-led with modest researcher control with/without manipulation checks
		No information	No information provided
	Fidelity measure[a]		Monitor, expert observation, self-report
Comparator		Passive control	(A) For acute studies = no intervention (e.g., resting/sitting condition without a prescribed additional task) (B) For chronic studies = Business as usual, waiting list, that is, any usual routine activity without an expectancy match (e.g., usual daily practice, school routine, PE delivered by generalists who did not receive specific training/instructions), solving academic tasks during curricular lessons
		Nonphysically active control	Sedentary activities with an expectancy match, that is, sedentary activities beyond daily routine of similar length, conducted by the same specifically trained persons as in the intervention groups, for example, (A) for acute studies: reading, watching educational videos, sitting in nature, solving academic tasks, relaxation techniques; (B) for chronic studies: participation to workshops or targeted sitting educational activities
		Physically active control	Alternative physical activity with an expectancy match, that is, intervention of similar length and conducted by the same category of specifically trained persons as in the intervention group (e.g., stretching, physical activity types, intensities and frequencies different from the intervention, PE delivered by researchers, external specialists, or specifically trained classroom teachers.
Design		Acute	Single PA session
		Chronic	Multiple PA sessions over at least 4 weeks

Note. MVPA = moderate-to-vigorous physical activity; MPA = moderate physical activity; PA = physical activity; PE = physical education; VPA = vigorous physical activity.
[a] Included in the review but not in the meta-analyses.

cognitive stage of motor learning); (c) motor–cognitive dual-task (DT) training, such as exergaming with variable perceptual–motor adaptation demands; (d) sports at the open skill end of the closed–open skills continuum (ball games, martial arts) with mostly high time pressure.

- Hybrid: (a) activities not explicitly designed to include, but possibly involving some degree/phases of cognitive challenge, and/or coordinative complexity, and/or DT training; (b) sports in between the closed and open skill continuum, or closed skill sports without an explicit focus on coordinative task complexity and without a high time pressure. Typical cases of hybrid activities are spurious types of activities such as multicomponent training, physical education, or holistic movement practices (yoga) without features of high coordinative complexity or time pressure.

Individual-Level Moderators

Personal or specific characteristics of the participants were coded.

Age. Preschool children (3–5 years), children (6–12 years), adolescents (13–17 years), adults (18–59 years), and older adults (60+ years) were included.

Special Focal Population. As suggested by Burnette et al. (2023, p. 10), "the focal group moderator combines at risk and other indicators of a targeted sample for which effects are expected to be pronounced—that is, individuals for whom the intervention is expected to be most efficacious." In our meta-analysis, we considered "focal group" individuals who are healthy and have not been diagnosed with a disease but may still have conditions that render them at risk of poor cognition and therefore potentially higher responders to physical activity interventions. We have encompassed nonpathologic conditions that range from genetic (e.g., apolipoprotein E gene) and developmental (e.g., low fitness, poor motor skills at baseline; postmenopausal age, prefrail, at risk of fall) to metabolic and/or behavioral (overweight), or environmental (vulnerable children living in disadvantaged environments; older adults living in care facilities).

Study-Level Moderators

This refers to the characteristics of the study and the implementation of the intervention.

Chronic/Acute. Chronic studies considered the longer lasting effects of an intervention with repeated physical activity bouts (of at least 4 weeks), whereas acute studies considered the transient effects of a single bout of physical activity.

Design. Chronic studies were categorized as RCT versus cluster RCT, whereas acute studies were categorized as RCT versus randomized within-subjects crossover pre–posttest trials versus randomized within-subjects crossover post-only trials.

Comparator (Type of Control). Studies were categorized depending on whether they employed, as control group, usual practice or active control. A control group was identified if it was defined by authors as such (control or comparator group); compared to the intervention group, it had to be less physically and/or cognitively demanding. According to Ziv et al. (2024), the identification of active versus passive controls was based on the extent to which the control condition involved the assignment of a task (physically or nonphysically active) that could generate similar expectations to improve performance for the experimental and control groups. Passive control: a condition where participants are not given a sedentary or physically active task to complete during the study period and are engaged in usual sedentary activities (e.g., waitlist for chronic studies or resting for acute studies). Nonphysically active control: a condition where participants are asked to engage in a sedentary cognitive task during the study period (e.g., engaging in learning tasks or watching educational videos). Active control: a condition where participants are asked to engage in a physically active task during the study period (e.g., a low-intensity activity such as stretching or a coordinative low-challenging activity such as simple walking or running).

Implementation Fidelity (Adherence and Compliance). These categories were added, with respect to the PROSPERO protocol, because the selected studies varied largely in the extent to which implementation procedures were described and implementation checks used. According to clinical literature (e.g., Mir, 2023, p. 219), "adherence is an active choice to follow through with the prescribed treatment, whereas compliance is a (relatively) passive behavior in which a patient is following a list of instructions." In chronic physical activity research, we can get information on both adherence to the overall intervention period (i.e., based on session attendance, as predominantly measured; Giovanazzi et al., 2022) and compliance with the intensity and type of PA at the single session level. Instead, in acute research, only compliance can be considered. We operationalized compliance in terms of "content control" (Burnette et al., 2023): (a) full researcher control with manipulation checks, (b) researcher-led without manipulation checks, (c) specialist-led with modest researcher control with/without manipulation checks. We operationalized adherence in terms of "attendance" (Giovanazzi et al., 2022): (a) as prescribed, with an average attendance rate of at least 80% of sessions; (b) somewhat as prescribed, with average attendance rates lower than 80%; (c) no information, if no information on attendance rates is provided, or very large range (e.g., 10%–90%) without an indication of a prevalent attendance range. In general, research has found that RCTs in public health in adults (Armstrong et al., 2008), complex interventions (e.g., physical activity; Bates, 2022), and education research in children (in particular, active learning; Vazou et al., 2020), do not consistently report process evaluation measures, making it hard to assess whether these programs had any deviations from the protocols.

Outcome Variables

According to the categorization used by Pontifex et al. (2019), cognitive outcomes included assessments of:

1. *Attention* involves conscious and controlled processing, delimiting divided attention to the extent to which different sources of information can be processed at the same time (Johnston & Dark, 1986). In this category, selective attention (e.g., d2 test), vigilance, and concentration are also included.

2. *Executive function* is an umbrella term used to identify core and higher level cognitive functions involved in the control of goal-directed behaviors (Diamond, 2013). The three core skills that are recognized as classical subcomponents

of executive function are (a) inhibition, the ability to stay focused and resist temptations; (b) cognitive flexibility, the ability to switch the focus of attention easily and flexibly; and (c) working memory, ability to "maintain selected information in an active, easily retrievable state while inhibiting distractors and interference (i.e., short-term memory and interference control at the attentional and cognitive levels)," or in simpler terms, "holding information in mind and working with it while not perceptually present" (Diamond, 2013, p.147).

3. *Memory* relates to any state or process resulting from the sequential stages of encoding, storage, and retrieval tasks (Klein, 2015), comprised of different types such as short-term, visuospatial, and episodic memory (e.g., recall, recognition tasks). This category excluded working memory, which was included as an executive function task.

4. *Information processing speed* (e.g., Letter Digit Substitution Test).

We further included:

5. *Creativity* "involves the development of a novel product, idea, or problem solution that is of value to the individual and/or the larger social group" (Hennessey & Amabile, 2010, p. 572).

6. *Global cognitive function* included other measures of cognitive function such as high-order cognitive function.

7. *Intelligence* (e.g., Raven's Progressive Matrices, Wide Range Achievement Test, Wechsler Intelligence Scales for Children).

Following Diamond (2013), we further summarized cognitive outcomes in the following categories:

1. *Core executive functions* (pure or more global, impure tests of the three core executive function tasks; inhibition, working memory, and cognitive flexibility), self-regulation.

2. *Other nonexecutive cognitive functions* (all speeded reaction time tasks without an executive function task component, including information processing speed in simple, discriminative, and choice reaction time tasks), attention (alerting, vigilance, concentration, selective, divided attention, visuospatial attention), and memory (short- and long-term memory in the different modalities—visual–verbal, episodic etc., excluding working memory that falls within the core executive function category).

3. *Higher level executive functions* (planning, reasoning, decision-making), intelligence and creativity.

Data Extraction

Data from included studies were extracted independently by two reviewers. The following key study characteristics were extracted: study location (country of origin) and year conducted, sample size, study design, age of participants and category, special focal population, program characteristics and comparators, physical activity intensity (type, relative/absolute, measure) and type, program duration, acute/chronic studies, cognitive demand, physical and social environment, experimental setting and domain, delivery mode and style, implementation fidelity (adherence, compliance, measure), cognitive outcomes, and study results. In the cases of disagreement, both reviewers undertook the data extraction and discrepancies (e.g., inclusion/exclusion of studies due to unclarity in reporting program components and/or comparators) were solved upon discussion until consensus was reached (the interrater reliability score was 92%).

Risk of Bias (Quality) Assessment

Two reviewers assessed the quality and risk of bias (RoB) for each of the included studies using the Revised Cochrane risk-of-bias tools for (a) randomized controlled and (b) crossover trials. A third reviewer was consulted in the case of discrepancies (the interrater reliability score was 88%). Studies with multiple outcomes were assessed separately. Each study was assessed against the following criteria: (a) randomization process, (b) deviations from intended interventions, (c) missing outcome data, (d) measurement of the outcomes, and (e) selection of the reported results. Signaling questions within each domain will have the following response options: "not applicable," "yes/probably yes," "no/probably no," and "no information," and will then be used to generate domain-specific judgments of either "low risk," "some concerns," or "high risk" of bias. An overall bias was calculated for each journal article assigned either "low risk," "some concerns," or "high risk." Studies were rated low RoB if all five domains were judged to be low risk, some concerns if at least one domain had some concern regarding bias, and high risk if at least one domain was judged to be high risk.

Strategy for Data Synthesis

All analyses were conducted using the metaSEM package (Cheung, 2015) in R Version 4.0.2. Pretest and posttest mean values and standard deviations from the intervention and control groups were extracted and converted to effect sizes (ESs), computed as the difference value between the mean pre–post change in the intervention and that in the control group, divided by the pooled standard deviation of the pre–post change value of the two groups.[2]

The computed Hedges' g was further bias-corrected as per Hedges and Olkin's (1986) small-study bias correction factor. We report Hedges' g which corrects for small sample bias instead of Cohen's d (with small sample sizes, $n < 20$). All scores were converted to Hedges' g effect size and were defined as small ($g = 0.20$), medium ($g = 0.50$), and large ($g = 0.80$; Cohen, 1988). A positive g indicates that the experimental group, only or more pronouncedly, improved cognitive performance compared to the control group (i.e., positive pre–post change in the experimental

[2] $g = \frac{(x_1 - x_2)}{\sqrt{\frac{((n_1-1)\times s_1^2 + (n_1-1)\times s_2^2)}{(n_1+n_2-2)}}}$

where
x_1 = mean pre–post change value for the intervention group
x_2 = mean pre–post change value for the control group
n_1 = subsample size of the intervention group
n_2 = subsample size of the control group
s_1 = standard deviation of pre–post change values for the intervention group
s_2 = standard deviation of pre–post change values for the control group

group vs. less pronounced, absent, or negative pre–post change in the control group, where "positive" refers to the meaning of cognitive test outcomes accounting for whether a higher or lower outcome value corresponds to better performance). Conversely, a negative *g* indicates a pre-to-post change more advantageous for the control group. Thus, positive *g* values mean that physical activity improved cognitive outcomes (and negative the reverse).

For chronic studies with multiple assessments during and after a longer intervention period (e.g., 2 years; Pesce, Lakes, et al., 2021), only the assessment matching the end of the intervention period was used. For acute studies with multiple assessment time points, we included all the time points reported. Where multiple tests of cognition were reported, we extracted each test score and treated them individually. When multiple measures of the same outcome were reported, we included the most meaningful and representative of the test (e.g., interference scores, switch costs, Trail Making Test Part B but not Trail Making Test Part A, Flanker incongruent but not congruent, Stroop colors not words). Studies dividing the sample into subgroups (e.g., low vs. high fitness level; low vs. high impulsivity) were included if they reported the mean, *SD*, and sample size of each subgroup (not only the overall sample size; e.g., Chu et al., 2015).

We performed both preliminary and main analyses separately for acute and chronic studies. The preliminary analyses include only primary studies with RCT design and nonphysically active control group to exclude the potential RoB that may arise when including studies that do not apply the gold standard of RCTs and, employing passive controls, might have inflated risk of larger effects because of expectancy of improvement in the intervention group regardless of the intervention features (i.e., Hawthorne effect; Román-Caballero et al., 2023). The main analyses, including all selected primary studies with any type of control groups, but accounting for it as a potential moderator, aimed to address the primary and exploratory research questions on the moderating role of contextual factors and their interplay with physical activity characteristics on the effects of chronic and acute physical activity on cognition.

Preliminary Analyses

We performed the following series of preliminary analyses, including all primary studies and only primary studies with RCT design and nonphysically active control group.[3]

1. Analysis of the overall physical activity effect on cognition: a three-level univariate meta-analysis with maximum likelihood estimation method, including the pretest standardized mean difference to account for baseline differences (for chronic studies only) and sampling variance (for both chronic and acute studies).

2. Search for outliers and sensitivity analysis: systematic search for outliers utilizing studentized residuals, Cook's Distance, difference in beta values, and predicted values (Viechtbauer & Cheung, 2010), and analysis of overall physical activity effect on cognition after removing the identified outliers.

3. Analysis for testing small studies effects/risk of publication bias: M. Egger et al., 1997; Higgins et al., 2023) and Bayesian Random Effects Meta-Analyses with Publication Selection Model-Averaging (Bartoš et al., 2022). This latter combines precision effect test and precision effect estimate with standard errors with selection models into one method that bases conclusions on multiple models simultaneously depending on how well they account for the data. This analysis was performed both maintaining and excluding outliers, identified as indicated in (2).

Main Meta-Analyses With Focus on Moderators Acting on the Physical Activity–Cognition Relationship

Main meta-analyses, including all primary studies, were aimed to address the primary and exploratory hypotheses on the role of moderators. For all meta-analyses, we combined effect sizes using a multilevel structural equation modeling approach. The main advantage of this approach is that it is not limited by the assumption of independence, and multiple effect sizes can be included from each study (i.e., effect sizes are nested within studies; Cheung, 2015). Unconditional mixed-effects models using maximum likelihood estimation were conducted to calculate the overall pooled effect size. For each pooled effect size, 95% likelihood-based confidence intervals were calculated. The I^2 statistic measures variability in the effect sizes (i.e., heterogeneity) explored in moderator analyses (Higgins et al., 2003) and is considered not important (0%–40%), moderate (30%–60%), substantial (50%–90%), and considerable (75%–100%; Reeves et al., 2008). A Q test as an omnibus test of the model and its goodness of fit was run.

For the analysis of chronic physical activity effects, we only included studies with pre- and posttest values and included in the analysis the pretest standardized mean difference to account for baseline differences. For the analysis of acute physical activity effects, we also included posttest-only studies, therefore without accounting for baseline differences. We made this choice because adjusting for baseline in crossover trials may introduce bias and is therefore discouraged both by the Cochrane group (Elbourne et al., 2002; Higgins & Eldridge, 2023) and in the Consolidated Standards of Reporting Trials statement extension to randomized crossover trials (Dwan et al., 2019).

Included moderators were physical environment and experimental setting, social environment, delivery mode and style, cognitive demand, physical activity type and intensity, age, focal population, study design (for acute studies only), comparator type, implementation fidelity (compliance for both acute and chronic studies and adherence for chronic studies only), and RoB. Separate regressions on age, program duration for chronic studies, and session duration for acute studies (continuous predictors) were also performed.

The overall main results of chronic and acute studies on cognitive outcomes, from the meta-analysis and RoB assessments, were used to complete a Grading of Recommendations, Assessment, Development, and Evaluation certainty assessment (Hultcrantz et al., 2017). MFM, JB, AW, and GT qualitatively assessed RoB, consistency, and precision and gave a summary rating—low, moderate, or high certainty of evidence.

[3] The authors thank Georgia Salanti for her precious statistical advice in the final revision of the article.

Exploratory Analyses: Interactions of Context-Level Moderators (Physical Environment and Virtual Reality) With Task-Level Moderators (Physical Activity Intensity and Cognitive Demand)

The choice of moderator variables included in the exploratory interaction analyses was based on changes in cognitive capacity and potential underlying mechanisms. We tested the interaction of context-level moderators (i.e., physical environment and virtual reality) with task-level quantitative (i.e., physical activity intensity) and qualitative (i.e., cognitive demand of physical tasks) moderators. The physical environment may be the underlying mechanism for providing a "restoring" effect, physical activity quantity may reflect the capacity "boosting" effect, and physical activity quality the capacity "building" effect (Pesce et al., 2023). We also tested the interactions of the virtual reality with physical activity moderators to infer whether the mechanisms of immersion and presence (Howard, 2019) had different expressions at different physical activity intensities.

These exploratory analyses were performed separately for chronic and acute studies (see Table 2). Not all these interactions and not all combinations of levels of these categorical moderators could be tested, especially in the analysis of acute studies, due to the low amount and unbalanced distribution of studies across levels of moderator variables. For example, there were no acute studies in the virtual reality environment to test for its interactive role with physical activity intensity or cognitive load.

Finally, since the moderation of physical activity effects on cognition by physical activity intensity could be affected by differences in its measurement, we ran a further exploratory interaction analysis. In this latter, we analyzed the joint effect of physical activity intensity and absolute/relative intensity measure used. To be able to perform this analysis, we limited the intensity categories to the most represented ones (i.e., MVPA and moderate physical activity [MPA]).

Transparency and Openness

Our review was preregistered with the International Prospective Registry of Systematic Reviews (PROSPERO No: CRD4202236 9269), guided by the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (Moher et al., 2015; Shamseer et al., 2015), and additional recommendations for systematic meta-reviews (Hennessy et al., 2019; Page et al., 2021). We adhered to the meta-analytic reporting standards guidelines for meta-analytic reporting (Appelbaum et al., 2018). All meta-analytic data, analysis code, and research materials (including our coding scheme) are available (M. Mavilidi, 2024). We adhered to the Preferred Reporting Items for Systematic Review and Meta-Analysis 2020 guidelines for systematic reviews (Page et al., 2021).

The review methods were established prior to the conduct of the review. Deviations from the protocol were undertaken along the review process to deal with reviewers' requests. Unlike the original PROSPERO registration, quasiexperimental studies were excluded to enhance the comparability with major meta-analyses that followed the gold standard of RCTs, whereas the range of comparators was extended from sedentary controls only to encompass studies with physically active controls if explicitly defined as a control group by authors. This was done in consideration of major meta-analyses showing different ESs across comparator types and of an article published after our protocol registration (Ziv et al., 2024) on the issue of comparators in physical activity and cognition research. Academic achievement outcomes, though being included in the PROSPERO protocol, were excluded from the meta-analysis to avoid a disproportional presence (and influence on overall effects) of outcomes that are coupled exclusively with the school-age range and the school setting.

Results

Study Selection

Our search generated 16,515 possible records in the initial search, and 28 possible records were identified through other sources, meaning previous reviews and meta-analyses of this literature and extensive search of reference lists from obtained reports (see Preferred Reporting Items for Systematic Review and Meta-Analysis flow diagram of studies included presented in Supplemental File 2). After downloading citations into EndNote, duplicates were removed. In total, 1,197 publications were identified as possibly relevant to the inclusion criteria. Of the full-text articles reviewed, 239 were eligible for qualitative synthesis and included in the meta-analysis with full data. Where insufficient or graphic information was presented for relevant variables (e.g., mean change scores), authors in studies from 2014 and onward were contacted via email to provide the requested data. When relevant descriptive statistics were not available, studies were removed from the meta-analysis ($k = 104$).

Study Characteristics and Participants

A detailed table with descriptive characteristics per study and the summary of the data (i.e., name and number of studies per category) can be found in Supplemental File 3.

Studies were published between 1989 and 2023. The total sample size was 48,625 participants. Studies were conducted in 40 different countries from all continents, with the most represented countries being the United States and Canada ($k = 57$), Australia and New Zealand ($k = 20$), Europe and the United Kingdom ($k = 94$), South America ($k = 13$), Africa ($k = 5$), and Asia ($k = 49$).

The study population included children ($k = 80$), adolescents ($k = 32$), adults ($k = 38$), and older adults ($k = 89$). Studies with special focal populations ($k = 46$) included overweight/obese ($k = 16$), narrowed to fitness level ($k = 6$), disadvantaged/low socioeconomic status/low performing ($k = 4$), not autonomously living adults (e.g., nursing/aged care homes, institutionalized; $k = 9$), and individuals at risk (e.g., falling, cognitive decline; $k = 9$). Mean age ranged between 4.2 and 84.8 years

Study Design and Quality and Implementation Fidelity

Acute/Chronic and Study Design. Most studies looked at the chronic effects of physical activity on cognitive outcomes ($k = 171$) and less at the acute effects ($k = 68$). Most studies (acute and chronic; $k = 209$) were RCTs and cluster RCTs with between-subjects design ($k = 165$ and $k = 44$, respectively) and less randomized crossover trials with within-subjects design (acute; $k = 30$).

Comparator Group. Most studies ($k = 149$) used a control condition that was sedentary passive (acute studies) or usual practice

Table 2

Summary of Main Findings, Including Effects, Number of Studies, Confidence Intervals, for the Joint Effects (i.e., Interactions) Between Context- and Physical Activity-Level Moderators of the Physical Activity–Cognition

Joint moderator	*k*	No. of ES	*M* g	[95% CI] *LL*	[95% CI] *UL*	I^2	τ_2^2	τ_3^2	R_2^2	R_3^2	*Q*	*df*(*Q*)	*p*
					Chronic studies								
Physical Environment × PA Intensity						62.6	0.01	0.07	0.68	0.11	16.52	3	.001*
Indoors—MVPA	64	249	**0.15**	**0.08**	**0.23**								
Indoors—MPA	46	146	**0.26**	**0.17**	**0.35**								
Outdoors—MVPA	9	46	**0.63**	**0.41**	**0.85**								
Outdoors—MPA	6	17	0.05	−0.21	0.31								
Physical Environment × Cognitive Demand						76.6	0.03	0.11	0.38	0.00	28.42	3	<.001*
Indoors—high	51	145	**0.20**	**0.09**	**0.31**								
Indoors—hybrid	40	108	**0.25**	**0.13**	**0.38**								
Outdoors—high	4	15	**0.53**	**0.15**	**0.91**								
Outdoors—hybrid	4	6	**1.01**	**0.51**	**1.52**								
PA Intensity × PA Intensity Assessment (relative/absolute)						68.1	0.00	0.07	0.63	0.15	13.34	3	.004*
MVPA—absolute	23	99	0.09	−0.04	0.22								
MVPA—relative	45	162	**0.25**	**0.16**	**0.34**								
MPA—absolute	5	19	0.22	−0.05	0.48								
MPA—relative	27	83	**0.26**	**0.15**	**0.38**								
					Acute studies								
Physical Environment × PA Intensity						41.9	0.01	0.04	0.00	0.35	38.46	3	<.001*
Indoors—MVPA	23	107	**0.19**	**0.08**	**0.29**								
Indoors—MPA	25	88	**0.21**	**0.10**	**0.31**								
Outdoors—MVPA	3	18	0.03	−0.20	0.26								
Outdoors—MPA	3	13	**0.71**	**0.42**	**0.99**								
Physical Environment × Cognitive Demand						59.0	0.01	0.07	0.02	0.02	0.53	2	.766
Indoors—high	16	50	0.21	0.05	0.40								
Indoors—hybrid	7	27	0.19	−0.02	0.40								
Outdoors—high	1	0											
Outdoors—hybrid	1	7	0.37	−0.04	0.80								
PA Intensity × PA Intensity Assessment (relative/absolute)						45.0	0.00	0.06	0.00	0.04	5.00	3	.172
MVPA—absolute	5	33	0.12	−0.12	0.36								
MVPA—relative	19	85	0.19	0.05	0.33								
MPA—absolute	3	11	0.28	−0.05	0.60								
MPA—relative	19	74	0.24	0.10	0.85								

Note. Values in bold represent effect sizes with confidence intervals that reached significance (95% CI did not include zero). Positive *g* values mean physical activity improved cognitive outcomes (and negative the reverse). CI = confidence intervals; *LL* = lower limit; *UL* = upper limit; ES = effect size; I^2 = % of heterogeneity; *M g* = Mean Hedges' *g*, where values are positive to the extent that physical activity increases cognitive outcomes; MVPA = moderate-to-vigorous physical activity; MPA = moderate physical activity; VPA = vigorous physical activity; PA = physical activity; τ_2^2 = within study variance; τ_3^2 = between study variance; R_2^2 = variance explained at Level 2 (i.e., between effect sizes from the same study); R_3^2 = variance explained at Level 3 (i.e., between studies); *Q* = omnibus test of heterogeneity; *df*(*Q*) = degrees of freedom of omnibus test.

* $p < .05$.

of daily living (i.e., "business as usual"; chronic studies); others used a nonphysically active ($k = 64$) or physically active control ($k = 30$).

RoB. Most studies were characterized as with "some concerns" ($k = 165$) or "high risk" ($k = 43$). Few studies were characterized as "low risk" ($k = 31$; see Supplemental File 4).

Implementation Fidelity (Adherence and Compliance). *Adherence* (for chronic studies only): Studies did not report information ($k = 73$), were somewhat as prescribed ($k = 33$) or as prescribed ($k = 67$). *Compliance* (for both chronic and acute studies): Studies had full ($k = 88$), partial ($k = 82$), and modest researcher control ($k = 56$). Few studies did not report any information ($k = 13$).

Cognitive Outcomes

A variety of measures for cognitive outcomes were utilized, including core executive functions ($k = 183$), higher order executive functions, intelligence and creativity ($k = 23$), and other nonexecutive cognitive functions ($k = 103$). Cognitive assessments used are summarized in Supplemental File 5.

Physical Activity Quantity and Quality

Physical Activity Quantity: Intensity and Program Duration. Most studies used MVPA ($k = 122$), moderate ($k = 90$), light ($k = 29$), vigorous ($k = 19$), or high/near-maximal physical activity ($k = 3$). Studies used an absolute ($k = 37$) and relative ($k = 105$) measure of physical activity intensity, both ($k = 16$), or did not provide this information ($k = 81$). Studies assessed physical activity intensity using an objective measure ($k = 108$), a subjective rating ($k = 18$), or both ($k = 33$), or intensity was estimated by expert judgment ($k = 75$) or was not reported ($k = 6$).

The duration of a single physical activity bout ranged from 5 to 90 min. The overall duration of chronic physical activity interventions ranged from 4 weeks up to 104 weeks.

Physical Activity Quality: Type and Cognitive Demand. Physical activity type included aerobic ($k = 98$), resistance training ($k = 23$), coordination training ($k = 20$), multicomponent ($k = 58$), motor–cognitive activities (including exergaming)/sports games ($k = 63$), or holistic movement practices/martial arts ($k = 24$). Studies involved high ($k = 95$), hybrid ($k = 63$), or low cognitive demand ($k = 123$).

Context: Environment, Setting/Domain, and Delivery

Physical and Social Environment. Studies were conducted indoors ($k = 180$), outdoors ($k = 23$), in a combination of places ($k = 18$), or the physical environment was not reported ($k = 22$). In most studies, physical activity was conducted in a group ($k = 181$); in fewer studies individually ($k = 50$). Few studies included both individual and group sessions ($k = 9$).

Experimental Setting and Domain. Settings included lab/university ($k = 62$), school ($k = 89$), community ($k = 78$), home, or a combination of settings ($k = 10$). Domains comprised school ($k = 80$), occupational ($k = 2$), and leisure ($k = 157$).

Delivery Mode and Style. Most interventions were delivered in face-to-face mode ($k = 200$), remote/virtual ($k = 21$), or combined mode ($k = 20$). As concerns the delivery style, most studies were atheoretical ($k = 202$), whereas a minority was based on a theoretical ($k = 14$) or evidence-based ($k = 23$) approach.

Preliminary Analyses

Overall Physical Activity Effects on Cognition

The three-level univariate meta-analysis with maximum likelihood estimation method performed with all studies, including the pretest standardized mean difference (for chronic studies only) and the sampling variance (for both chronic and acute studies). An overall ES $g = 0.25$, 95% CI [0.19, 0.31] was found for chronic studies and 0.21, 95% CI [0.12, 0.30] for acute studies. When including only RCTs with nonphysically active controls, the overall ES was $g = 0.21$, 95% CI [0.11, 0.31] for chronic studies and $g = 0.15$, 95% CI [0.00, 0.31] for acute studies (see also Figure 2).

Search for Outliers and Sensitivity Analysis

The studentized residuals, Cook's Distance, difference in beta values, and hat values showed that there were six chronic studies and six acute studies that were potential outliers or influential studies. After excluding them, the ES was $g = 0.20$, 95% CI [0.16, 0.25] for all chronic studies and $g = 0.15$, 95% CI [0.07, 0.23] for all acute studies. After excluding these potential outliers and narrowing the analysis only on RCTs with nonphysically active controls, the ESs and CIs remained unchanged as in the analysis without outliers' removal ($g = 0.21$, 95% CI [0.11, 0.31] for chronic studies and $g = 0.15$, 95% CI [0.00, 0.31] for acute studies), meaning that there were no outliers in the ESs of the subgroup of studies with nonphysically active controls.

Testing for Small Studies Effects/Risk of Publication Bias

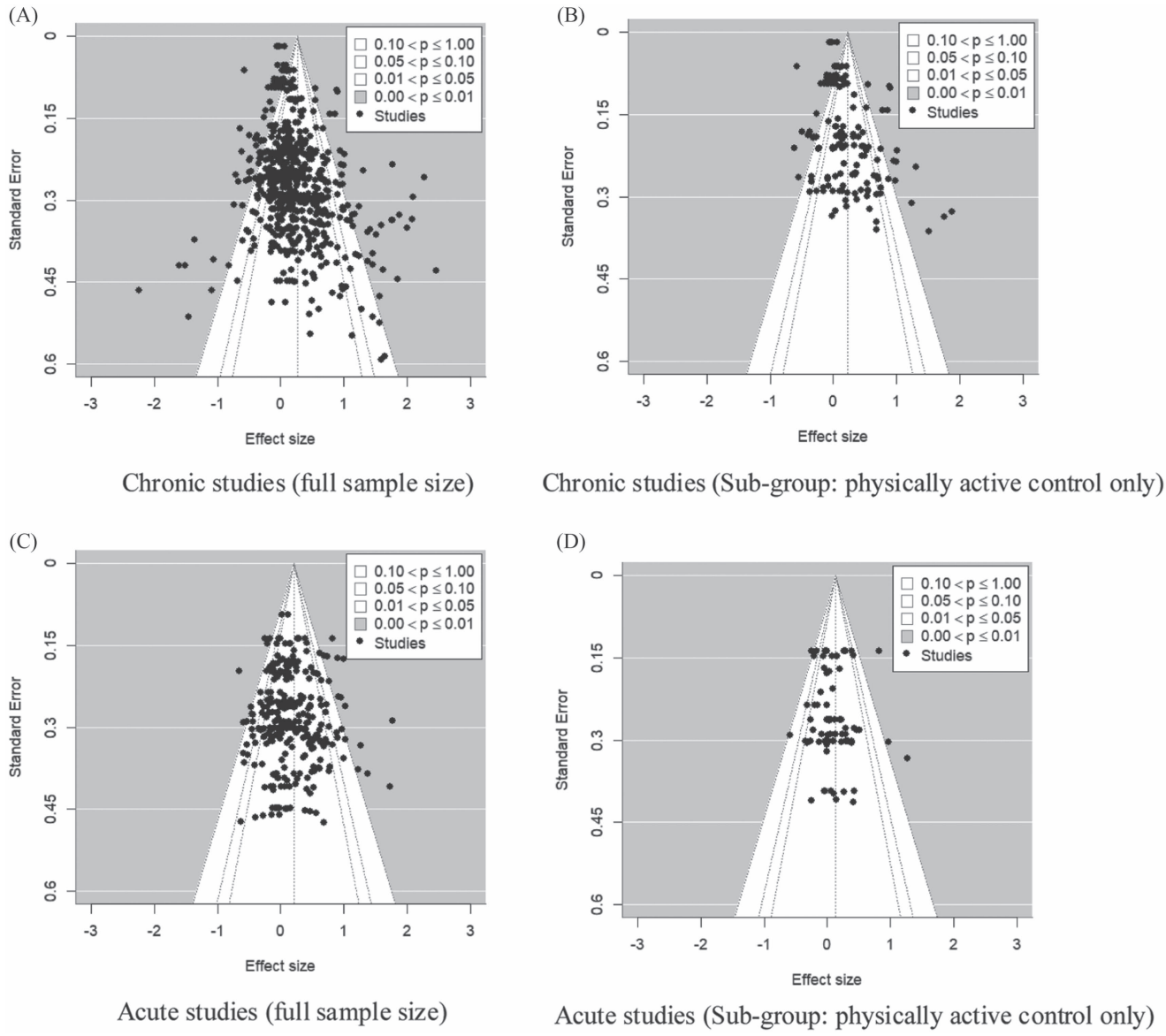
Results from the Bayesian Random Effects Meta-Analyses with Publication Selection Model-Averaging (Bartoš et al., 2022) did not differ when maintaining or excluding potential outliers. Strong evidence of larger effects in smaller studies was found among all chronic studies ($BF_{10} = 657.7$; Egger's test: $p < .0001$), as well as in the subsample of chronic RCTs with nonphysically active controls ($BF_{10} = 48.3$; Egger's test: $p < .0001$), with a model-average ES of 0.01, 95% CI [−0.01, 0.11] and 0.01, 95% CI [−0.01, 0.18], respectively.

Some evidence of larger effects in smaller studies emerged among all acute studies, ($BF_{10} = 0.42$; Egger's test: $p = .51$), with a model-average ES of 0.18, 95% CI [0.00, 0.29], and strong evidence of larger effects in smaller studies ($BF_{10} = 6.3$; Egger's test: $p = .94$) emerged in the subsample of acute studies with nonphysically active controls, with a model-average ES of 0.00, 95% CI [−0.19, 0.18].

Main Meta-Analyses With Focus on Moderators Acting on the Physical Activity–Cognition Relationship

Contour-enhanced funnel plots are displayed in Figure 2 for chronic and acute studies. The results of the meta-analyses performed, separately for chronic and acute studies, to address the role of individual moderators of the physical activity–cognition relation are depicted in Figure 3 and presented in Supplemental File 6 as follows: (a) overall physical activity effects on cognition, (b) effects of context-level, (c) task-level, (d) individual- and study-level moderators. The results of the meta-analyses performed to explore the joint effects (i.e., interactions) of context-level and physical activity-level moderators are displayed in Figure 4 and presented in

Figure 2
Overall Effects of Physical Activity on Cognition



Note. Funnel plots. A positive g indicates that the experimental group, only or more pronouncedly, improved in cognitive performance (i.e., pre-to-post change) compared to the control group. Thus, positive *g* values mean that physical activity improved cognitive outcomes (and negative the reverse).

Table 2. The certainty of evidence for the effects of chronic and acute studies on cognitive outcomes is displayed in Table 3.

Overall Effects of Physical Activity on Cognition

In chronic studies, physical activity had a small positive effect on cognition (g = 0.25, 95% CI [0.19, 0.31], k = 171) with considerable heterogeneity (I^2 = 89.8%). No significant differences were observed between specific cognitive outcomes. In acute studies, physical activity also had a small effect on cognition (g = 0.21, 95% CI [0.12, 0.30], k = 68), with substantial heterogeneity (I^2 = 55,2%) and significant differences across outcome types ($p < .001$). Core executive function (g = 0.16, 95% CI [0.08, 0.24]), higher level executive function (g = 0.28, 95% CI [0.18, 0.39]), and nonexecutive cognitive function (g = 0.32, 95% CI [0.16, 0.48]) had small ES with CI not including zero.

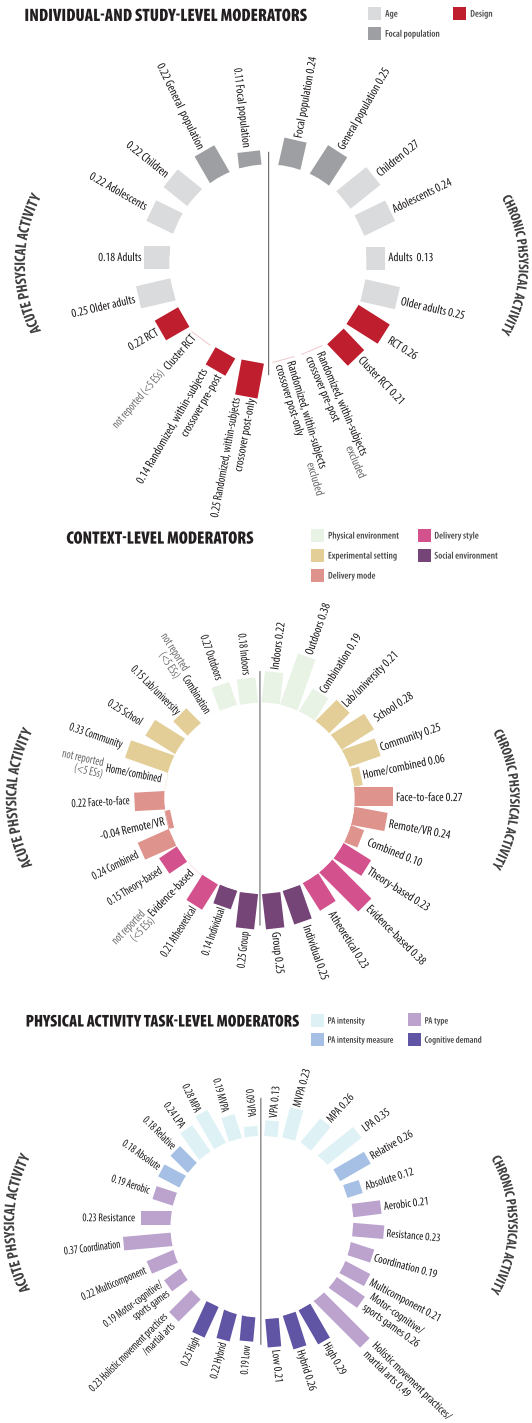
Effects of Context-Level Moderators on the Physical Activity–Cognition Relationship

Physical and Social Environment, Experimental Setting. No significant differences emerged for chronic or acute studies, between the small-size effects of these moderators.

Delivery Mode and Style. No significant differences emerged, for both chronic and acute studies in delivery modes. Significant differences (p = .011) emerged among delivery styles for chronic

Figure 3

Effect Sizes of Moderators Acting on the Physical Activity–Cognition Relationship



Note. RCT = randomized controlled trials; ES = effect size; PA = physical activity; LPA = light PA; MPA = moderate PA; MVPA = moderate-to-vigorous PA; VPA = vigorous PA; VR = virtual reality. See the online article for the color version of this figure.

studies, with larger ES for evidence-based ($g = 0.38$, 95% CI [0.22, 0.55]) compared to theory-based ($g = 0.23$, 95% CI [0.01, 0.45]) or atheoretical styles ($g = 0.23$, 95% CI [0.17, 0.30]).

Effects of Physical Activity-Level Moderators

Quantitative Physical Activity Characteristics. Regarding physical activity intensity, no significant differences in ES among intensity levels were found for chronic studies. In acute studies, significant differences emerged among intensity levels ($p = .001$). Light physical activity ($g = 0.24$, 95% CI [0.04, 0.40]), moderate physical activity (MPA; $g = 0.28$, 95% CI [0.17, 0.38]), and MVPA ($g = 0.19$, 95% CI [0.08, 0.29]) had small ESs with CI not including the zero, whereas vigorous physical activity/near-maximal intensity had a negligible effect ($g = 0.09$, 95% CI [0.04, 0.22]). Regarding the relative/absolute assessment of physical activity intensity, a significant difference was found for chronic studies ($p = .012$), with larger ES for relative ($g = 0.26$, 95% CI [0.17, 0.34]) compared to absolute assessment ($g = 0.12$, 95% CI [−0.01, 0.26]), but not for acute studies. The prediction of cognitive performance accrued by program duration (chronic studies) and session duration (acute studies) was negligible.

Qualitative Physical Activity Characteristics. Significant differences across physical activity types emerged for chronic studies ($p = .001$, $g = 0.19$–0.49), with the largest ES found for holistic movement practices/martial arts ($g = 0.49$, 95% CI [0.17, 0.66]), followed by motor cognitive/sports games ($g = 0.26$, 95% CI [0.17, 0.35]). In acute studies, an only marginally significant ($p = .059$) moderation by physical activity type was likely due to the largest ES for coordination training ($g = 0.37$). Small but significant differences across levels of cognitive demand ($p = .005$) emerged for chronic studies, with $g = 0.21$, 95% CI [0.13, 0.30], 0.26, 95% CI [0.18, 0.34], and 0.29, 95% CI [0.22, 0.36] for low, hybrid, and high cognitive demand, respectively, but not for acute studies ($p = .099$, $g = 0.19$–0.25).

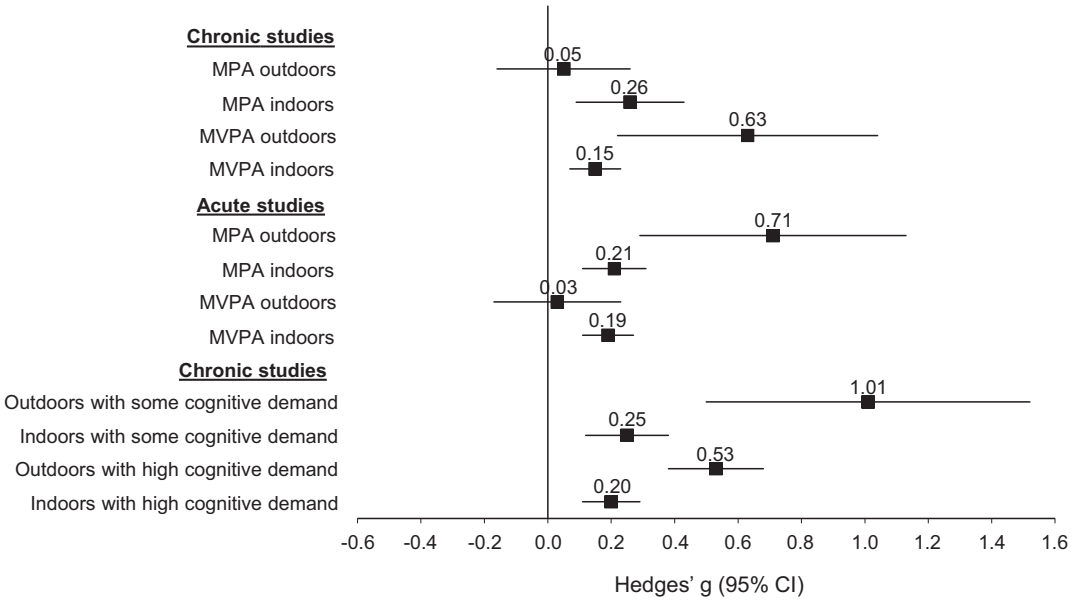
Effects of Individual- and Study-Level Moderators

Age, Special Focal Population, Design, and Control Type. No significant differences in ES were found across age classes neither for chronic nor for acute studies. The prediction by age was negligible in the meta-regression performed with the discrete age variable. No differences were shown between general and special focal population for chronic or acute studies. Instead, the study design significantly impacted ES only in acute studies ($p = .026$), with small ESs found for RCT ($g = 0.22$, 95% CI [0.12, 0.31]) and randomized within-subjects crossover posttest-only studies ($g = 0.25$, 95% CI [0.09, 0.41]), but not for randomized, within-subjects crossover pre–posttest studies. Regarding the comparator type, a significant difference emerged for chronic studies ($p = .022$), with small effects found for passive ($g = 0.29$, 95% CI [0.23, 0.36]) and nonphysically active controls ($g = 0.21$, 95% CI [0.06, 0.35]), but not for physically active controls. No comparator-type effect was found for acute studies.

Implementation Fidelity (Adherence and Compliance) and RoB. For chronic studies, there was neither an effect of adherence (i.e., attendance rate whenever assessed) nor an effect of

Figure 4

Effects Sizes of Exploratory Analyses: Task-Level Moderators (Physical Activity Intensity) and Interactions of Context-Level Moderators (Physical Environment) With Task-Level Moderators (Cognitive Demand)



Note. MVPA = moderate-to-vigorous physical activity; MPA = moderate physical activity; CI = confidence interval.

compliance. Instead, there was a significant effect of compliance for acute studies ($p = .002$), with modest ($g = 0.39$, 95% CI [0.14, 0.65]) and partial researcher control ($g = 0.29$, 95% CI [0.15, 0.42]) showing larger ESs compared to full researcher control ($g = 0.15$, 95% CI [0.06, 0.27]). Regarding the RoB, chronic studies with some concern showed a significantly ($g = 0.29$, 95% CI [0.22, 0.37], $p = .17$) larger effect than studies with high ($g = 0.18$, 95% CI [0.06, 0.30]) or low concern ($g = 0.18$, 95% CI [0.04, 0.32]). No effects of RoB were found for acute studies.

Joint Effects of Context- and Physical Activity-Level Moderators

Physical Environment × Physical Activity Intensity. Significant interaction effects were found for both chronic ($p = .001$) and acute studies ($p < .001$). In chronic studies, those with MVPA conducted outdoors had the largest effect (of moderate size, $g = 0.63$, 95% CI [0.41, 0.85]) that was three times larger than the ES of interventions of the same MVPA intensity conducted indoors ($g = 0.15$, 95% CI [0.08, 0.23]). In acute studies, the advantage of the outdoor environment emerged for interventions of MPA intensity, with studies conducted outdoors having an effect (of moderate size, $g = 0.71$, 95% CI [0.42, 0.99]) larger than the ES of interventions of the same MPA intensity conducted indoors ($g = 0.21$, 95% CI [0.10, 0.31]).

Physical Environment × Cognitive Demand. Significant interaction effects were found for chronic studies only ($p < .001$). Interventions conducted outdoors had large effects ($g = 1.01$, 95% CI [0.52, 1.52], from six ESs only) when the physical activity had some degree of cognitive demand (i.e., hybrid) and moderate effects when the cognitive demand to perform the physical activity was high ($g = 0.53$, 95% CI [0.15, 0.91]). Interventions with the corresponding levels of cognitive demand but conducted indoors yielded only small effects ($g = 0.25$, 95% CI [0.13, 0.38] and $g = 0.20$, 95% CI [0.09, 0.31] for indoor interventions with hybrid and high cognitive demand, respectively).

Virtual Reality × Physical Activity Intensity/Cognitive Demand. Only chronic studies provided sufficient ESs for running the planned interaction analysis of Virtual Reality × Physical Activity Intensity, which did not yield significant results. Neither

Table 3

Certainty of Evidence for the Overall Effects of Physical Activity on Cognitive Outcomes

Type of study and result	*k* of studies	*n* of participants	*M* g [95% CI]	Certainty of evidence
Chronic studies				
Overall effects of physical activity on cognitive outcome	171	11,327	0.25 [0.19, 0.31]	Moderate certainty
Acute studies				
Overall effects of physical activity on cognitive outcomes	68	3,501	0.21 [0.12, 0.30]	Moderate certainty

Note. CI = confidence intervals. *M g* = Mean Hedges' *g*, where values are positive to the extent that physical activity improved cognitive outcomes.

chronic, nor acute studies provided sufficient ESs for contrasting virtual reality physical activity with hybrid versus high level of cognitive demand.

Physical Activity Intensity × Physical Activity Intensity Assessment (Relative/Absolute). Significant effects were found for chronic ($p = .004$) but not for acute studies. For chronic studies, interventions with MVPA or MPA intensity had a meaningful ES (i.e., with CI crossing the zero) if the intensity was assessed with a relative measure (MVPA: $g = 0.25$, 95% CI [0.16, 0.34]; MPA: $g = 0.26$, 95% CI [0.15, 0.38]), but not if the intensity assessment measure was absolute.

Discussion

This comprehensive review aimed at evaluating (a) the moderating effect of contextual factors on cognitive outcomes of physical activity and (b) their joint influence with more commonly investigated moderators of the physical activity task (quantity and quality). To achieve our goal, we separately analyzed chronic and acute physical activity interventions with participants of any age, from children of 3 years to older adults.

Overall, we observed a small positive effect of both chronic and acute physical activity on cognition, with a strong risk of publication bias for chronic studies and some risk for acute studies. At first glance, none of the considered context-level factors except delivery style, with the largest effects for evidence-based interventions, seem to influence the size of these small effects. Instead, the type of physical activity seems to matter, with the largest chronic physical activity effects for holistic movement practices and the largest transient effects of single bouts of physical activity for coordination training. Most interestingly, the context and the quantity and quality of physical activity seem to influence the size of the effects interactively.

The overall small positive effects of both chronic ($g = 0.25$) and acute physical activity ($g = 0.21$) on cognition are consistent with previous reviews of studies with chronic (Álvarez-Bueno et al., 2017; Ludyga et al., 2020) and acute physical activity design (Chang et al., 2012; Hillman et al., 2019; Ludyga et al., 2016; Pontifex et al., 2019). Instead, our small positive effect for chronic studies is in contrast with the negligible effect reported by Ciria et al. (2023), even though to maintain some methodological similarity, we included only RCTs, accounted for baseline differences, and, in a preliminary analysis, restricted the sample only to studies with a nonphysically active control. The most likely differences regard our less strict criteria to handle outliers and the risk of publication bias and our broader inclusion of healthy general and focal populations and physical activity interventions. The focus of the present review was not on the overall effects of physical activity on cognition but on the role of a range of moderators, including the qualitative characteristics of physical activity. To this aim, we also included physical activity interventions designed to entail increased cognitive demand and holistic movement practices (Vergeer & Biddle, 2021), which are reported as most beneficial to cognition (Diamond & Ling, 2020; Tomporowski & Pesce, 2019) and showed larger effects than other chronic intervention also in our meta-analysis.

The main effects of physical activity on cognition were also computed for each of the considered cognitive outcome domains: core and higher level executive functions and nonexecutive cognitive functions. Since all cognitive function types in both acute and chronic studies yielded small-size effects, all with CI not including the zero, our results suggest that these effects are general, and the cognitive outcome domains can be combined to address the role of the contextual moderators and their eventual interplay with task-level moderators.

Similar to Pesce et al. (2023), our primary aim was to test the role of context in the physical activity–cognition relationship. We defined and analyzed different facets of the broader construct of context, including the physical environment (outdoors vs. indoors) and social environment (group vs. individual), the experimental setting (university/laboratory, school, community, home), the delivery mode (face-to-face, remote, virtual reality), and delivery style (theory-based vs. evidence-based vs. atheoretical). The absence of any influence of contextual factors, except delivery style, on the size of the overall little cognitive benefits of chronic and acute physical activity disconfirms our primary hypothesis.

The Physical Context Is Not Just a Background but an Interactive Ground

Regarding the *physical environment*, the presence of only non-significantly, slightly larger ESs for chronic and acute outdoor physical activity is in contrast with the hypothesis that added effects can be gained from performing physical activity in natural outdoor environments, also referred to as "green exercise" (Bamberg et al., 2018). However, this tendency to a cognitive advantage for physical activity practiced outdoors becomes larger and significant in conjunction with given quantitative and qualitative characteristics of physical activity: the intensity (for both chronic and acute studies) and the cognitive demand (for chronic studies only). Thus, the outdoor physical environment seems to be a contextual feature that can amplify the otherwise generally small-sized effect of physical activity on cognition only if coupled with the prolonged practice of physical activity of moderate-to-vigorous intensity ($g = 0.63$, 46 ESs) and with at least some or a high degree of cognitive demand ($g = 0.53$, 15 ESs, and $g = 1.01$, 7 ESs, respectively), or with a single session of moderate-intensity physical activity ($g = 0.71$, 13 ESs).

The cognitive benefits of exposure to green environments have been demonstrated in studies involving children and adolescents (Vella-Brodrick & Gilowska, 2022), adults, and older adults (Cassarino & Setti, 2015). According to the attention restoration theory (Kaplan, 1995), exposure to a natural landscape engages automatic attention processes allowing effortful, capacity-limited processes to be restored. Alternatively, the stress reduction theory (Ulrich et al., 1991) suggests that exposure to nature has stress-relief properties that trigger a sympathetic reaction, lessening the negative consequences of stress on cognition.

Most of the physical activity and cognition studies that have tested the above theories with interventions in the outdoor environment mainly focus on acute physical activity. Our results show an advantage of an outdoor physical activity bout when its intensity is moderate. 15 min of simple walking outdoors seem sufficient to improve the following cognitive performance and the brain activity that supports it, whereas 15 min of indoor walking do not exert any effect on the brain and cognition (e.g., Boere et al., 2023). The transient benefits of moderate-intensity green exercise, such as simply walking in nature, are observed in nonexecutive, core executive (Boere et al., 2023), and higher level executive function (creativity; Oppezzo & Schwartz, 2014). Instead, more sophisticated designs that manipulated the levels of "greenness" of the physical activity context but did not

deliberately manipulate or control the activity intensity yielded mixed results (Han, 2021; Wade et al., 2020).

There is extensive research on the greenspace–cognition relationship and on the broader physical, psychosocial, and mental health effects of green exercise for children and adolescents (e.g., Mnich et al., 2019) as well as adults (e.g., Hartig et al., 2014; Lahart et al., 2019; Thompson Coon et al., 2011) and older adults (Gagliardi & Piccinini, 2019). Still, studies at the intersection between these two research lines are surprisingly rare. The present meta-analysis of this kind of study suggests that to reap the largest benefits in the longer term, individuals should habitually engage outdoors in MVPA and/or physical activities that require at least some degree of cognitive engagement ($g = 1.01$). Nonetheless, we are cautious about the generalizability of this large cognitive benefit based on six ESs only. More reliably (15 ESs), the cognitive challenge in physical activity should be deliberately set high to obtain a cognitive benefit of moderate size ($g = 0.53$).

The results of our exploratory analysis may inform the design of future research on optimal configurations of contextual and physical activity characteristics. The cognitive capacity "boosting" effect of MVPA intensity and the capacity "building" effect of physical activities that are deliberately designed to generate cognitive challenge (Pesce et al., 2023) might benefit cognitive function most if practiced in the outdoor environment, which has properties that "restore" capacity-limited resources (Ohly et al., 2016; Stevenson et al., 2018) and reduce the impact of factors that impair cognition (Yao et al., 2021).

Although this meta-analysis was not powered for examining the interaction of the physical environment with the multiple categories of *physical activity type*, it is noted that the largest effect in chronic studies ($g = 0.49$) was found for holistic movement practices (Vergeer & Biddle, 2021) that are frequently practiced in outdoor natural setting (Djernis et al., 2019). This paves the way for future research to verify the potential additive or interactive effects of the natural context with the unique features of mindful physical activity. Moreover, more research is needed to investigate further characteristics of the outdoor environment, such as the degree of "greenness," exposure to pollution, and complexity of the "built" environment, which may influence the direction and size of the effects on cognition (Cassarino & Setti, 2015; Markevych et al., 2017). For example, physical activity outdoors with high air pollution or environmental stressors, such as cycling near high-traffic roads, may even turn into a negative influence and completely suppress the physical activity-related release of neurotrophic factors (e.g., Bos et al., 2011; You et al., 2022). In contrast, being active with a low concentration of pollutants may benefit cognition (e.g., Cassilhas et al., 2022).

Nonphysical Context Facets: Social Environment, Experimental Setting, and Delivery

We did not find any significant moderating effects of the *social environment*. The similarly small efficacy of group and individual delivery in chronic studies is consistent with the results of the meta-review by Pesce et al. (2023), which failed to confirm the hypothesis that social engagement and support may add to the cognitive benefits of physical effort. In the present meta-analysis, we could not distinguish the effect of physical activity participation within a group by choice, which is probably linked to more positive social interaction and support from the effect of physical activity participation within a group due to the experimental domain, like in a school or a classroom. Group physical activity may largely differ in the amount of social interaction: Fitness training groups likely involve lower and less complex interactions than team games. Chronic and acute studies that consider the presence or intensity of social interaction suggest that physical activity effects on cognition may be explained by social engagement (Robitaille et al., 2014; Zehnder et al., 2024). Yet, we did not find sufficient information on social interaction in the selected studies to analyze it as a moderator. The quality of social interactions within a group, such as group cohesion or perceptions of relatedness, should be examined in future studies to explore the social context.

Regarding the *experimental setting*, no significant differences emerged among laboratory/university, school, home, and community across all ages, likely due to covariations: The school setting is coupled with school age, and community-based interventions prevail in the older population. The minority of interventional studies conducted in leisure time physical activity settings (e.g., sports centers, home) did not confirm the noninterventional evidence on the predictive value of leisure physical activity for cognitive function maintenance until older adulthood (Engeroff et al., 2018).

Regarding the *delivery mode*, most acute and chronic programs were delivered face-to-face, and no differential effect emerged compared to remote or virtual reality conditions, in contrast to other meta-analytical evidence suggesting a moderate effect of prolonged practice of action video games on various aspects of cognition (Bediou et al., 2018). Thus, the mechanism of immersion and presence, as well as engagement and agency proposed to act in the virtual reality context (e.g., exergaming) and to underlie beneficial cognitive outcomes (Howard, 2019; Schuemie et al., 2001) seem not to bring added value compared to more conventional face-to-face delivery. Likely, in the exergaming field, manifold applications are available, which consequently also differ in their characteristics (Benzing & Schmidt, 2018; Stojan & Voelcker-Rehage, 2019). Moreover, a series of studies with children and adolescents consistently demonstrate that the benefits of acute bouts of exergaming are moderated by physical activity dose and cognitive demand (Anzeneder, Zehnder, Martin-Niedecken, et al., 2023; Anzeneder, Zehnder, Schmid, et al., 2023; Benzing et al., 2016). Unfortunately, the still limited number of acute exergaming studies that met our selection criteria has not allowed us to test the interactive role of these moderators. For example, chronic exergaming interventions seem efficacious with special populations such as attention-deficit/hyperactivity disorder (Benzing & Schmidt, 2019) or pediatric cancer survivors (Benzing et al., 2020), not included in this review.

Regarding the *delivery style*, we found an advantage, though limited to the small-size range ($g = 0.38$), for chronic interventions that are delivered in an evidence-based manner. These studies largely vary in physical activity content (e.g., from aerobic or resistance training to dual-task or multicomponent training and music-based multitasking); their sole commonality seems to be the fact that they refer to existing evidence of their efficacy. Yet, the frequent lack of information on the instructional style and the prevalence of interventions that are delivered atheoretically call for greater attention to the manipulation and reporting of delivery (Vella, Sutcliffe, et al., 2023) to identify the unique features that render evidence-based delivery more efficacious, as well as for more well-designed studies delivered using theoretically informed

pedagogical approaches (e.g., D. R. Lubans, Leahy, et al., 2021; Pesce, Ballester, & Benzing, 2021).

Implementation Fidelity, Adherence to Protocol, and RoB

Regarding *implementation fidelity,* in chronic studies, neither a higher attendance rate (*adherence*) nor a higher researcher control (*compliance*), seem to translate into larger cognitive benefits. These results are consistent with the finding that a low RoB is not predictive of larger beneficial effects on cognition. Instead, in acute studies, more positive effects emerged when the *compliance* to the rules and activities was modest or partial rather than full researcher control. Although physical activity programs implemented in laboratory and controlled research settings can be delivered with a high level of fidelity, they can lack ecological validity. Most studies in the education context insufficiently describe their intervention components and do not follow guidelines (Fynn et al., 2020), making it hard to decide on compliance. Also, differences in achieved intensity across studies may affect the implementation fidelity and adherence to protocol in both chronic (Erickson et al., 2019) and acute physical activity studies (Pontifex et al., 2019), as there is high heterogeneity in measures and thresholds used to define the intensity of the physical effort. Our categorization of intensity assessment as absolute (e.g., measured by accelerometers) or relative (e.g., accounting for the true individual range of cardiac capacity, as percentage of heart rate reserve) showed that ESs were higher and more reliable when physical activity intensity was defined and monitored in relative terms. Instead, the choice of the same physical activity intensity in absolute value for all participants may lead to different levels of internal load in different individuals, therefore enhancing the heterogeneity in individual responsiveness (Herold et al., 2019). Finally in acute research, studies with randomized, within-subjects crossover pre–posttest design showed the lowest, nonsignificant ES ($g = 0.25$), followed by RCTs ($g = 0.22$). This result is surprising because it is considered the most rigorous approach to acute physical activity research, incorporating the key attributes of between- and within-subjects designs.

This comprehensive review is the first that attempted to include potential contextual factors thought to moderate the relationship between physical activity and cognition. Its innovative design and number of studies included are a considerable strength and contribution to the field, which can be used as a guide for informing future research.

Limitations

Limitations should be acknowledged. Given the very large number of studies screened and included, and moderators coded, selection and data extraction were not fully performed in duplicate. Individual reviewers coded subsets of moderators, yet after achieving good agreement in the coding with a second reviewer. For the same reason, we did not compile a list of excluded studies but merely provided overall justifications for exclusion.

Considering the novelty of the topic, we addressed Level 1 moderators exhaustively, whereas we explored the effects of only a few selected combinations of physical activity context with quantitative and qualitative physical activity task characteristics (Level 2 moderators, see Table 2). This choice was partly made based on theorized mechanisms and partly to avoid too small number of studies in each variable, by aligning, with caution, to a minimum threshold of five (e.g., Ludyga et al., 2016) and possibly more than five ESs (Jackson & Turner, 2017). Nevertheless, underpowered subgroup analyses may have limited the generalization of the results (Cuijpers et al., 2021). For example, when the number of ESs was small (six), as in the case of the interaction between the outdoor environment and hybrid cognitive demand, a larger ES ($g = 1.01$) with disproportionally larger CIs emerged. The study questions focused on moderators and their interplay that requested to split studies into subgroups did not allow including only low RoB RCTs. This may have especially contributed to the heterogeneity of chronic study effects, since in the analysis with RoB as a moderator, chronic studies with some concern exhibited larger ESs.

We cannot exclude that outliers may have disproportionally contributed to the observed effects. Indeed, sensitivity analyses showed a reduction in ES from 0.25 to 0.20 and from 0.21 to 0.15 for chronic and acute studies, respectively, after removing six outliers each. We maintained those outliers in line with Viechtbauer and Cheung (2010) who did not advocate the routine deletion of outliers or influential studies but recommended using the appropriate statistics to detect them and estimate their influence as a part of sensitivity analyses. Moreover, specifically in physical activity research, studies identified as potential outliers with unusually large or small effects can point to characteristics of treatments or experimental conditions that produce these types of effects, thus leading to new insights about potential moderators (Viechtbauer & Cheung, 2010).

This is a relevant point for the aim of our meta-analysis, because our main questions and hypotheses regard the individual and joint role of moderators acting on the physical activity–cognition relation. For example, outliers might involve intervention features of mindful or cognitively challenging physical activity or exercise with high coordination demands that have shown largest ESs in previous meta-analyses (e.g., Diamond & Ling, 2020; Ludyga et al., 2020; see also meta-analyses presented synoptically in Tomporowski & Pesce, 2019, as de Greeff et al., 2018 for children and adolescents; or Gheysen et al., 2018 for older adults). However, visual inspection of the features of the studies with outliers neither confirmed this hypothesis, nor revealed a disproportional or exclusive concentration within a specific category of any other physical activity- or context-level moderator. Thus, outliers may signal the presence of moderator dimensions not considered in the present meta-analysis. Only for chronic studies, the six outliers were exclusively comprised in the category of passive control that was the most numerous by far ($n = 121$).

Importantly, some evidence consistent with risk of publication bias emerged for chronic studies more pronouncedly than for acute studies. This evidence calls for caution when drawing conclusions. However, we considered more appropriate to speak of "risk of publication bias" instead of "publication bias" (Afonso et al., 2024), as none of the available methods can estimate with sufficient certainty how many effect sizes are truly missing. Accordingly, we conceived the estimation of the risk of publication bias as a sensitivity analysis (Higgins et al., 2023) not necessarily to be followed by an adjustment to account for it. Rather, in line with Afonso et al.'s (2024) recommendations, we attempted to mitigate the risk of publication bias preventively by adopting recognized criteria: preregistration; search

strategy without restriction on date or language; inclusion of gray literature; request of data from authors; updating of the literature search. Caution in the interpretation in terms of publication bias is warranted, in the present meta-analysis, also considering the high heterogeneity, which implies that more than one population effect size exists against the assumption of a single population effect most publication bias tests rely upon (Johnson, 2021). In physical activity and cognition research, where nonsignificant intervention outcomes are commonly reported, confounds with sample size rather than missing publication of nonpredicted findings might have played a role in determining what looks like a publication bias. For example, moderators beyond those analyzed in this review may have acted as confounds in large, more than in small-sized studies, which are typically characterized by a stronger control for confounding and moderating factors.

Moreover, the ES was higher for chronic studies with passive controls ($g = 0.29$) than for active controls ($g = 0.11$–0.21). The RoB related to the choice of the control group is an issue of actual debate, which has been addressed statistically (Román-Caballero et al., 2023) or conceptually in terms of the expectancy value of the different experimental conditions (Ziv et al., 2024). Since our differences across groups were within the small ES range and the ES computed including only studies with nonphysically active controls was still significant ($g = 0.21$ for chronic and $g = 0.15$ for acute studies—see the Preliminary Analyses section), we maintained all types of control groups to avoid a loss of power for the moderator and interaction analyses. However, it cannot be excluded that a Hawthorne effect may have partially influenced the effects found for chronic but not acute studies.

Our data showed a substantial to considerable heterogeneity, but the results were not adjusted for heterogeneity, because we did not aim at testing overall effects of physical activity on cognition but the individual and joint role of a prespecified set of moderators that were expected to explain the presence of heterogeneity (Johnson, 2021). Also, heterogeneity might be partly due to the highly heterogeneous number of outcomes among studies, even though we included, when available, only difference scores (e.g., trail making Part B–Part A) or summary scores (e.g., Cognitive Assessment System subscale performances). A way to further dampen this problem is the use of overarching outcome constructs, obtained by averaging individual measures from similar tests. However, this would have reduced the discriminative power of our analyses, which showed differential effects for different cognitive outcome domains in acute studies.

In our review, we calculated ESs separately for chronic and acute studies to account for the partially different mechanisms underlying the transient and longer term effects of physical activity. In this way, some analyses were underpowered due to the small number of studies in each category. With the advancement of research with RCTs including contextual factors, future research will be able to give insight and more robust results on combinations of moderators of physical activity on cognition (i.e., Level 2 and Level 3 interactions, e.g., Cognitive Demand × Social Environment × Adults).

Due to the limited information available in the primary studies, the classification of the level of cognitive demand was conducted by expert judgment. Future research should identify and employ tools for quantifying cognitive engagement that derives from task complexity more objectively (O. Chen, Paas, & Sweller, 2023; Martin, 2018). Moreover, it is to consider that cognitive task complexity interacts with the individual skillfulness to generate a relative level of perceived cognitive challenge. Unfortunately, the level of cognitive challenge has been assessed only rarely by means of subjective reporting, such as a modified Borg scale (Anzeneder, Zehnder, Martin-Niedecken, et al., 2023; Anzeneder, Zehnder, Schmid, et al., 2023; Benzing et al., 2016; M. Schmidt et al., 2016).

It is important to note that the adult population ($n = 127$) was equally represented compared to children and adolescents ($n = 112$). The number of studies with "low" risk was substantially small, due to the lack of reporting and consistency in reporting protocols and process evaluation measures, which limit the possibility of inferring adherence to protocol with certainty. Moreover, very few studies report on the session quality (e.g., Supportive, Active, Autonomous, Fair, Enjoyable principles, D. R. Lubans, Leahy, et al., 2021). Future studies should aim to embed more qualitative information on process evaluation. If, on the one hand, information on implementation fidelity (adherence and compliance) is scarce, on the other hand, our evaluation of compliance and adherence is also limited by differences in the measurement of achieved intensity and cognitive demand across interventions. The included studies were not harmonized at the level of relative physical activity intensity and relative cognitive load. This heterogeneity may have negatively impacted our results, which showed that a lack of individualization of the physical activity intensity and cognitive challenge can lower the ESs.

Finally, in our analyses, we included only chronic RCT studies that provided both pre- and posttest values and acute RCT and randomized crossover studies with pre–posttest and post-only design. Future studies could also include nonexperimental designs (e.g., quasiexperimental), common in studies oriented toward a higher ecological validity.

Conclusions

The current review found that several facets of the physical activity context, including physical and social environment, domain, and delivery mode do not moderate the effects of physical activity on cognition individually. Instead, the outdoor physical environment seems to amplify the beneficial effect of physical activity of specific doses and features. Our findings show promise that providing people of all ages with opportunities to be active in natural outdoor environments can be conducive to enhanced cognitive functioning.

This meta-analysis was not performed to provide an exhaustive overview and draw definitive conclusions but to "upgrade" the physical activity context from the mere background of interventions to a moderator of physical activity effects on cognition. Our meta-analysis was limited by the fact that RCTs, which are widely considered the gold standard to address causality, often do not or insufficiently and subjectively report on relevant features of the implementation context (Pesce et al., 2023).

The results of our exploratory interaction analyses might inspire future integrative efforts and pave the way for investigating the joint role of contextual factors and more commonly studied characteristics of the physical activity intervention and the population. Our initial evidence synthesis calls for more research to investigate whether "green exercise" experiences across the lifespan, as those provided by outdoor skill education (Cottrell & Cottrell, 2020), may

help relieve cognitive overload and stress and benefit cognitive health. Future studies might target a joint analysis of nature gradients with levels of physical activity intensity or cognitive challenge to evaluate whether larger cognitive benefits of physical activity may be obtained in environments with stronger natural features (Wade et al., 2020), or with a stronger engagement with nature (Han, 2021). Also, given the relevance of active commuting in terms of sustainability and reduction of polluting emissions, more research should be performed in outdoor greenspaces, comparing their influence with that of urban outdoor environments. Further, studies embedding technology in the program component and delivery will help us infer whether it has a facilitating, detrimental, or no role in the physical activity–cognition relationship. Particular attention should be paid to executive functions, as they are most responsive to risk factors in the physical activity environment as pollution and stressors, as well as to protective and promoting factors that arise from the physical, socioemotional, and cognitive features of physical activity training.

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